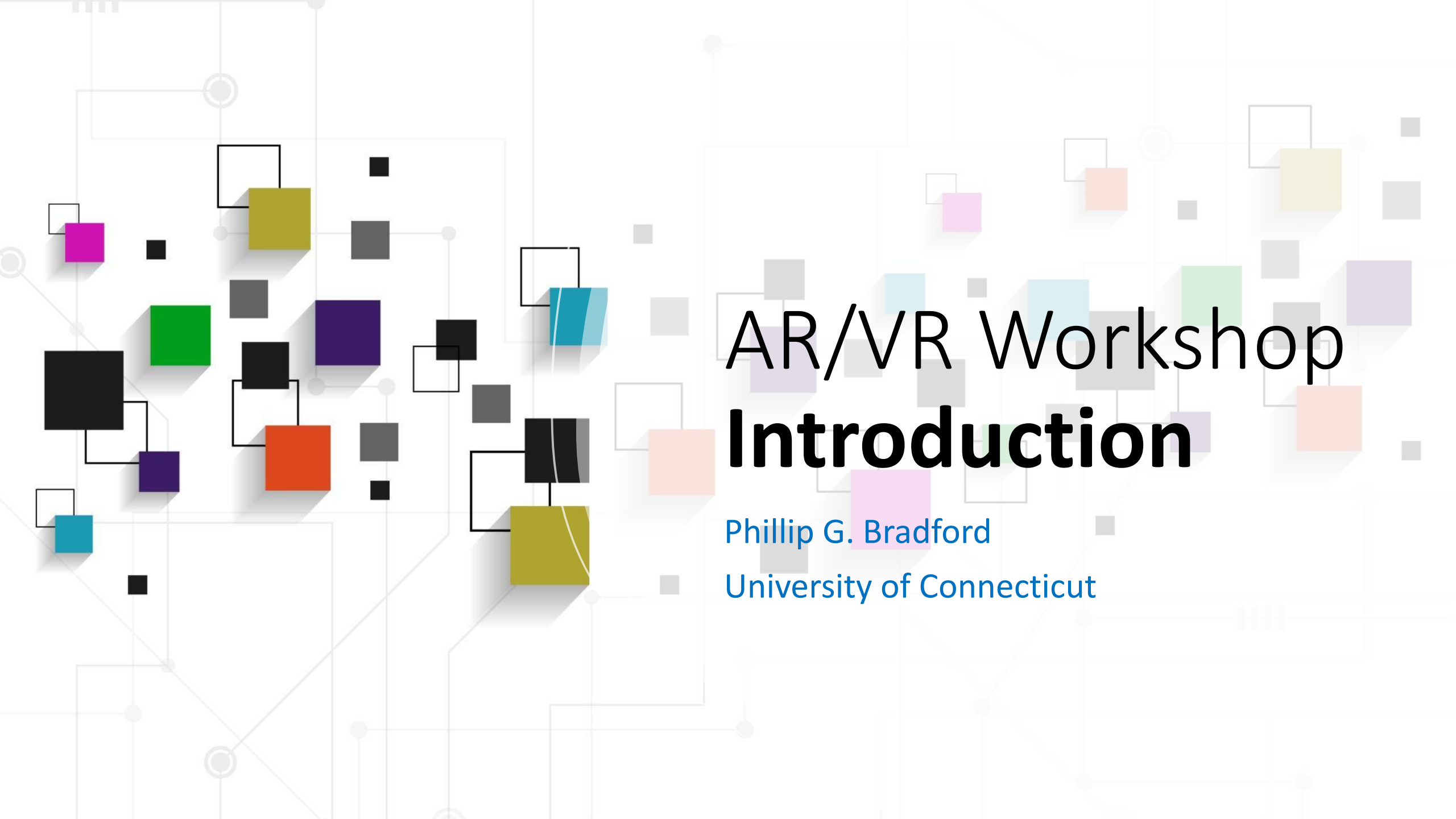


The background of the slide is a light gray grid with various colored squares (black, white, yellow, green, blue, red, purple, orange) and lines. Some squares have a 3D effect with a shadow. The title text is centered on the right side of the slide.

AR/VR Workshop Introduction

Phillip G. Bradford

University of Connecticut

Full select demo – how you can do ...



[Basic-Mars-Landing](#)

Module demo highlights: 3D image



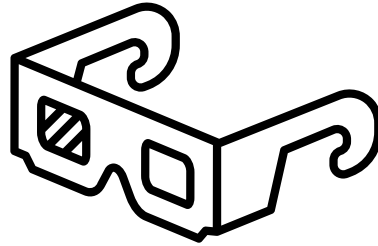
Outline

Overview

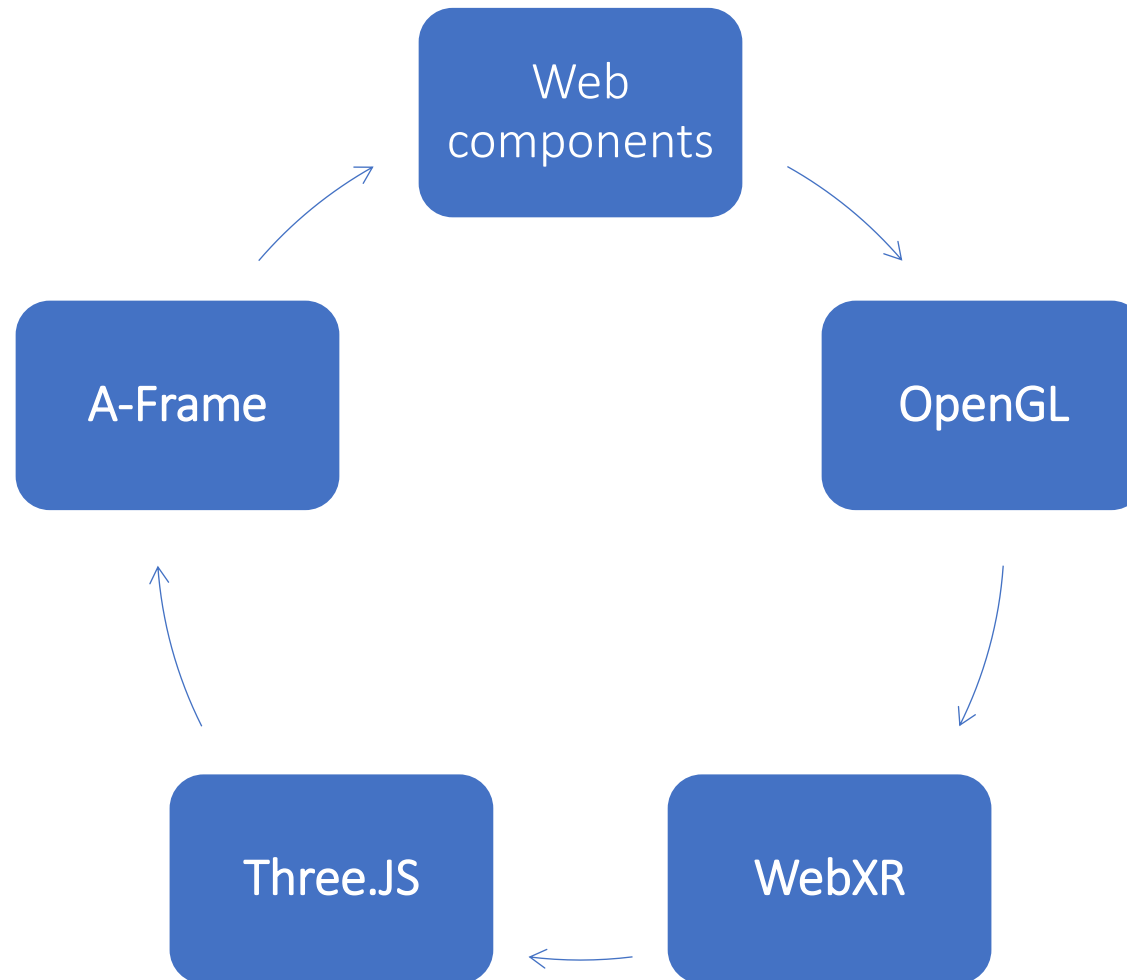
No installations needed

Github pages

Our first VR/AR examples



High level view and learning path



A-frame

3D virtual reality system built by Mozilla

A-frame is a component system built on **Three.js**

Three.js is built on **WebXL** which is built on **WebGL**

A-frame is HTML that uses web-components

Declarative – tell it what you want to see,
it figures out how to do it

No installation VR

Easy

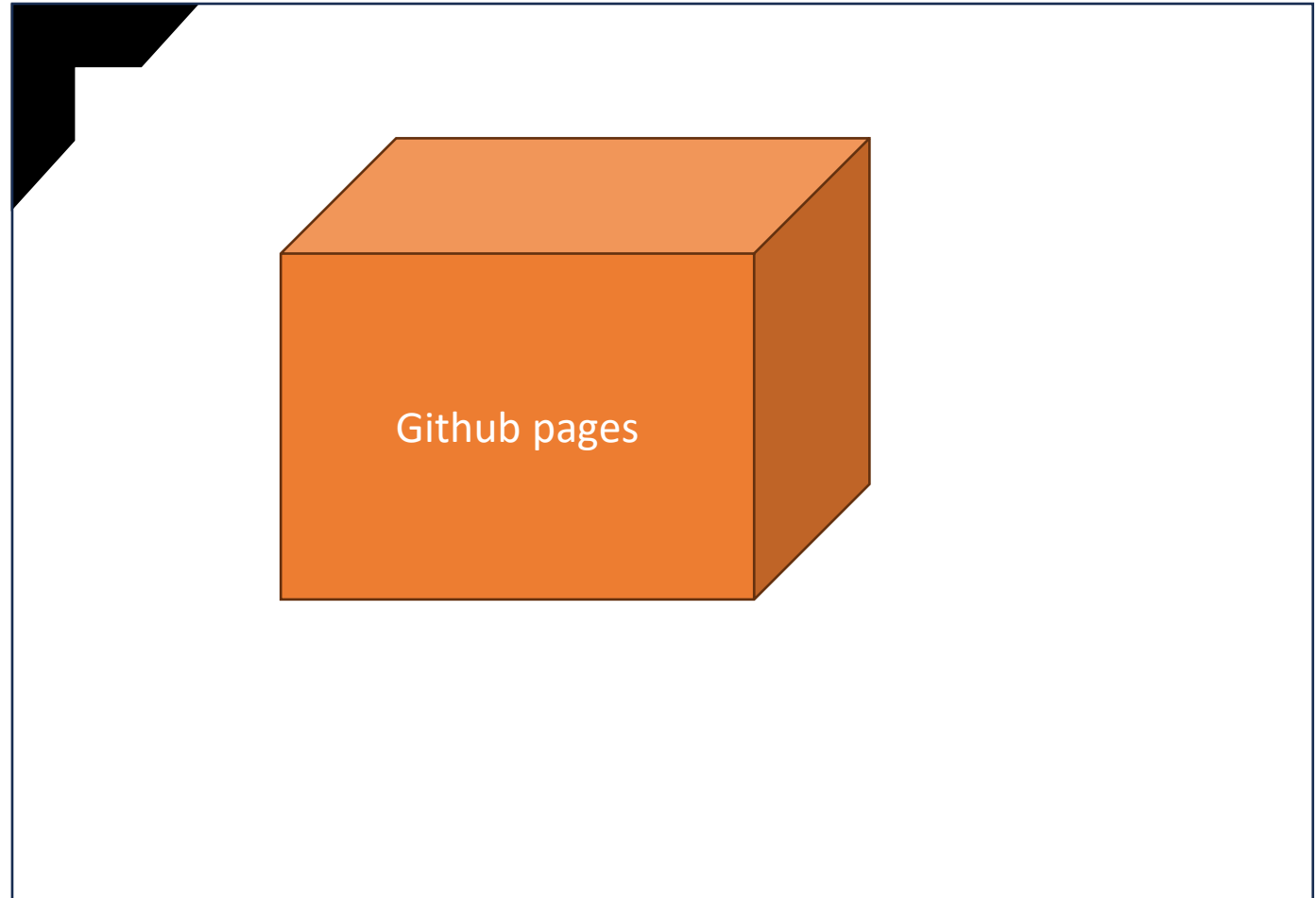
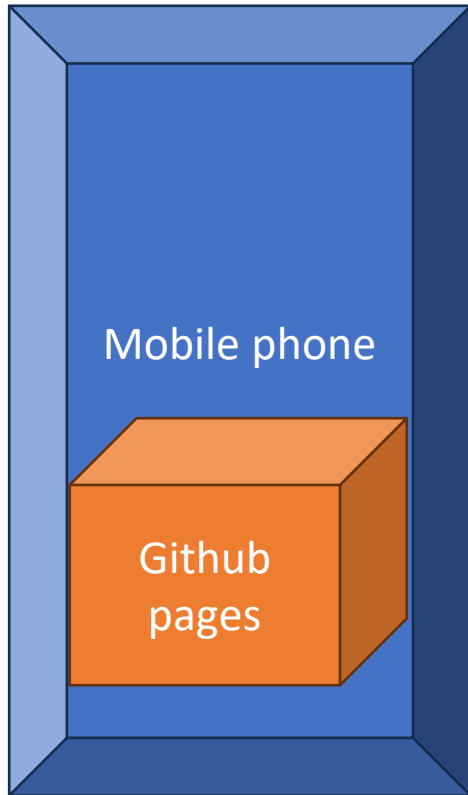
Only Python3 installed

Go to wonder-phil.github.io or YOUR_USER.github.io

Clone <https://github.com/wonder-phil/wonder-phil.github.io.git>

Overview: wonder-phil.github.io

Computer



Laptop view

```
git clone https://github.com/wonder-phil/wonder-phil.github.io.git
```

```
cd wonder-phil.github.io
```

```
Terminal> code .
```

```
Terminal> python3 -m http.server
```

In a browser go to <http://localhost:8000/>

Basic HTML

```
<html>
```

```
  <head>
```

```
    </head>
```

```
    <body>
```

```
      </body>
```

```
    </html>
```

index.html and not-index.html

```
<html>
<head>
  <meta charset="utf-8">
  <title>Hello A-Frame Example</title>
  <script src="https://aframe.io/releases/1.7.1/aframe.min.js"></script>
</head>
<body>
  <a-scene>
    <a-sky color="#0000EE"></a-sky>
    <a-text
      font="kelsonsans"
      color = "#FFFFFF"
      value="HELLO A-Frame!!!"
      width="6"
      position="0 -1 -5"
    ></a-text>
  </a-scene>
</body>
</html>
```

<a-scene>

```
<a-sky color="#0000EE">
</a-sky>
<a-text
  font="kelsonsans"
  color = "#FFFFFF"
  value="HELLO A-Frame!!!"
  width="6"
  position="0 -1 -5"
></a-text>
</a-scene>
```

VR glasses steps

On your phone go to your github pages

wonder-phil.github.io

Use VR glasses to view the project on your phone

simple-sky.html

<a-scene>

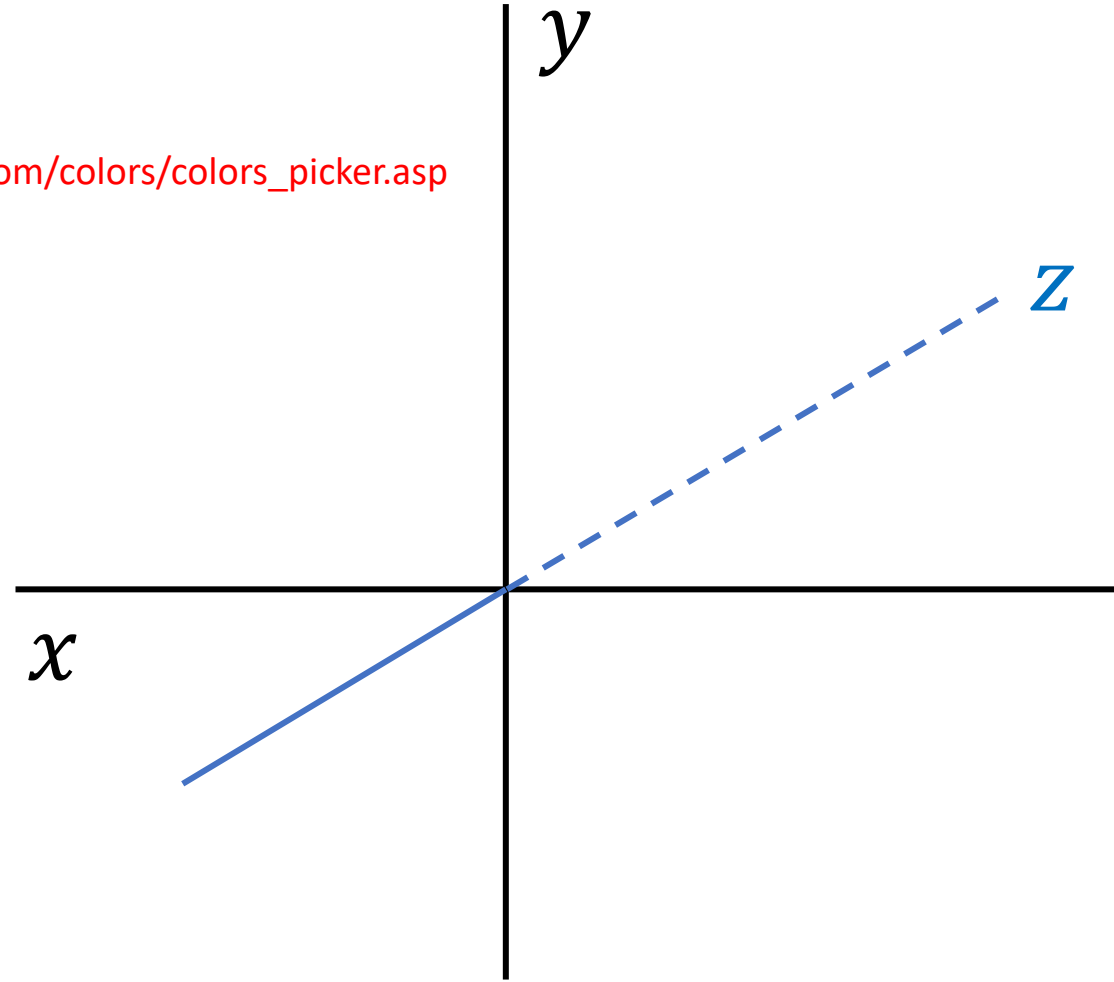
<a-sky color="#0105EF"> Select a color at https://www.w3schools.com/colors/colors_picker.asp
</a-sky>

<a-text
 font="kelsonsans"
 color = "#FFFFFF"
 value="HELLO A-Frame!!!"
 width="6"

- position="0 -1 -5" Note: **x-horizontal, y-vertical, and z-depth***
></a-text>

</a-scene>

*this translates to 1 meter below the origin and 5 meters into the page



simple-sky.html: basic color

```
<a-scene>  
  <a-sky color="#0105EF">  
  </a-sky>  
  <a-text  
    font="kelsonsans"  
    color = "#FFFFFF"  
    value="HELLO A-Frame!!!"  
    width="6"  
    position="0  -1  -5"  
  ></a-text>  
</a-scene>
```

Replace the current index.html on glitch
with not-index.html
Simple-sky.html

← a-scene now looks like this

simple-sky.html: changing the view

```
<script>
  const src = 'http://localhost:8000/p01/IMAGES/heic2007a.jpg'
  const img = document.getElementById('nebula');
  img.setAttribute('src', src);    // set it
  document.querySelector('a-assets').addEventListener('loaded', () => {
    document.getElementById('mySky').setAttribute('material', 'src: #nebula');
  });
</script>
<a-scene>
  <a-assets>
    <img id="nebula" crossorigin="anonymous">
  </a-assets>

  <a-sky id="mySky"
    src="#nebula" position="-2.5 0 -1.5"
  ></a-sky>
  <a-text
    font="kelsonsans"
    color = "#FFFFFF"
    value="HELLO A-Frame from EARTH!!!"
    width="6"
    position="0 -1 -5"
  ></a-text>
</a-scene>
```

Putting it all together

You have now created virtual reality text on a 3D background

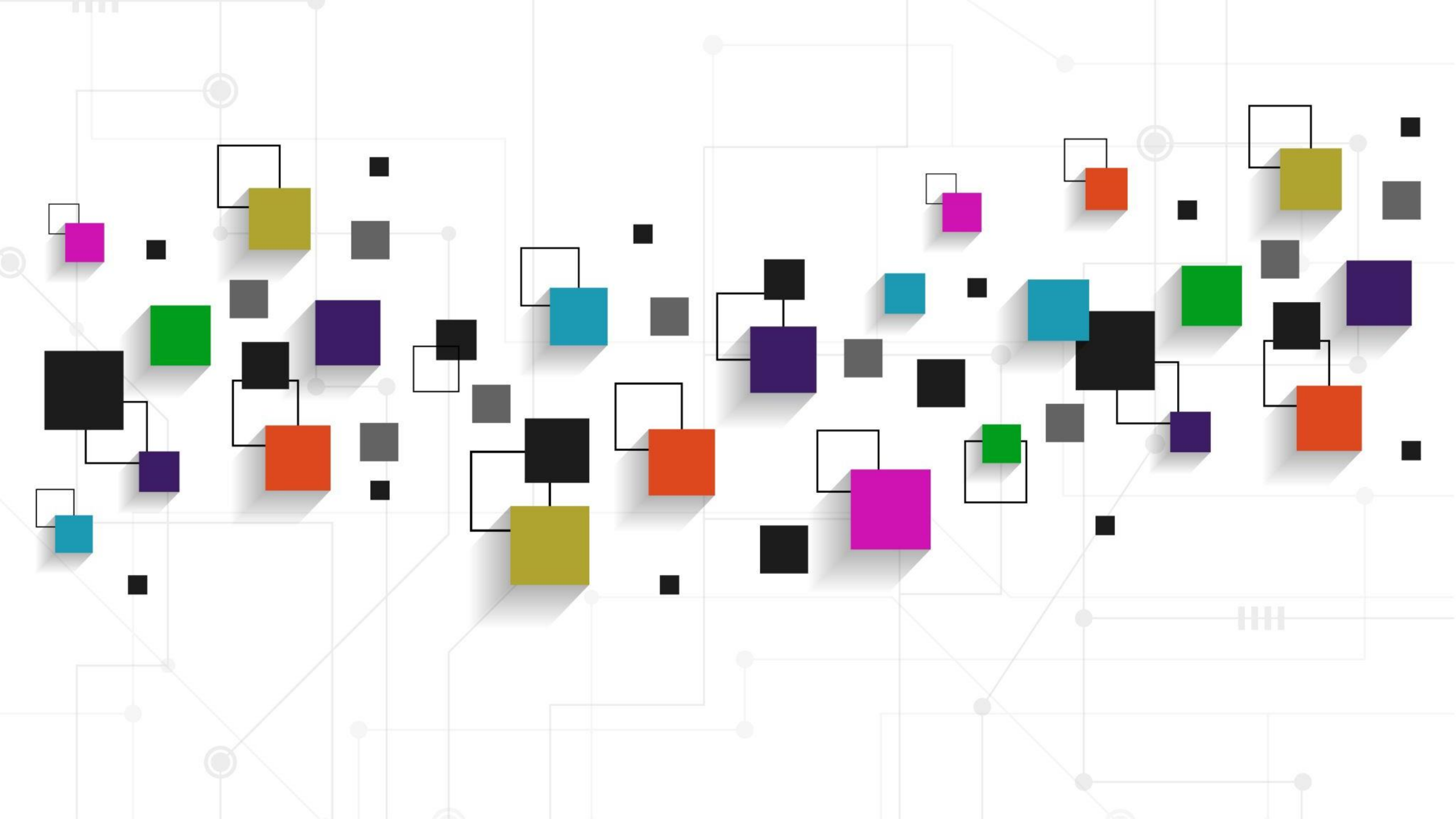
We can also

1. Change background – to the nebula

What if it doesn't work? TROUBLESHOOTING!

2. Move text around in the nebula

What if it doesn't work? TROUBLESHOOTING!



Getting a-frame from aframe.io

```
<head>  
  <script src="https://aframe.io/releases/1.7.1/aframe.min.js">  
  </script>  
</head>
```

Or to run outside of header to help TROUBLE SHOOTING

```
<body>  
  <script src="../../js/aframe-master.min.js"> <!-- OLD a-frame here -->  
                                          <!-- NEW a-frame in head -->  
  </script>
```

TROUBLESHOOTING!

CTRL-ALT-I in browser for a-frame debugging

CMD-Option-I in browser for a-frame debugging

SHIFT-CTRL-J in browser for Chrome/FireFox debugging

CTRL-O in browser to load a file